THE SPACE SHUTTLE IN PERSPECTIVE +

*S. Neil Hosenball**

Brief Description of the Shuttle

The Space Shuttle flight system is composed of the Orbiter—an external tank and two solid rocket boosters. The Orbiter has a cargo bay in which the Spacelab or other payloads may be carried. The Shuttle may carry propulsion stages or upper stages, which may be used to take satellites from Earth orbit into either synchronous orbits, circular orbits, geostationary orbits or planetary trajectories to one of the outer planets or to the moon.

The Space Shuttle is a reusable launch vehicle. It can deploy satellites and undertake experiments; it can do on-orbit servicing and retrieve payloads. It is only an earth-orbiting vehicle. It can go from an orbit of about 100 nautical miles to about 600 nautical miles above the Earth. It cannot go further into outer space or deep space. It can be used either as a space laboratory with a Spacelab or for other types of experimentation, even without the Spacelab on board. It will be able to carry up to 65,000 lbs. in its 60 x 15 ft. cargo bay. Its use can substantially decrease the cost per flight. For instance, by the use of the Shuttle, the cost of a delta payload used for launching a communications satellite could be reduced to about half the cost.

With this truncated description of what the Shuttle is and what it can do, let me limit my perspective of the Space Shuttle to its commercial aspects.

Charges to Users

One of the concerns of the fairly large number of commercial users of communications satellites (RCA, AT&T, Western Union, INTELSAT) has been that they were charged actual costs. Consequently, they would sometimes receive a bill for additional costs, far exceeding the original estimate—either as a result of inflation, strikes, a lower than predicted launch rate or some other reason causing a price increase. Commercial users prefer to have a fixed price for obvious reasons; future planning and fixing their cost exposure being just two of many good reasons. The Space Shuttle is a considerable improvement on this situation, inasmuch as all standard launch services charges, with the exception of escalation due to inflation, are included in the fixed price. In addition, there is a re-flight guarantee to the effect that if a mission is aborted, it will be flown again for free. In other words, there will not be any charge for the second mission if the abort was not caused through fault of a commercial user. In case of an expendable launch vehicle, if there were a failure for any reason whatsoever, including NASA's fault, one had to buy a new launch.

---

*General Counsel, NASA.

+ *Editor's note:* This article the text of which has recently appeared in "The Space Shuttle and the Law", edited by Professor Stephen Gorove is reproduced here with permission and exceptionally, for the reason that it formed an integral part of the Symposium on "Space Law in Perspective".

The user of the Shuttle may get 20% standby discount if the user is willing to go when he is told to go. This is similar to the practice of airlines giving discount on a standby basis. The user may also get discounts on charter flights when they buy the whole payload capacity. The share-user concept is quite evident with respect to the Shuttle.

Schedule of Launches and Reimbursement

Originally, when the Space Shuttle program was first suggested, nobody was concerned about NASA's meeting its schedule. Instead, the concern was whether there would be sufficient payloads to fully utilize the Shuttle's capabilities. It was thought that NASA would probably meet its schedule, just like it met its Apollo schedule. Now there is concern about the delay, since there would be considerable cost-saving for commercial operations, if the Shuttle were used instead of an expendable launch vehicle. It has been estimated that there will be sixteen launches in 1982, seven of which will be NASA launches, while the others will be U.S. commercial, foreign or military launches. In subsequent years, the number of commercial and foreign launches is expected to increase. A significant number of these launches will be reimbursed launches. At the current level of expendables, the U.S. is being reimbursed approximately two to three-hundred million dollars a year, for either vehicles or services that are being rendered for foreign government and U.S. commercial customers.

Kinds of Payload and their Selection

If there is some extra space onboard, some very small self-contained payloads, so-called getaway specials, may be thrown on for a cheap price. However, rather rigid rules apply as to what can be placed in them. Many universities have bought a small self-contained payload for experiments; several groups have bought some for high school students. The Shuttle will enable us to send up such payloads into space relatively cheaply and provide an opportunity to both non-profit and private enterprises to experiment at small cost. The Shuttle is not going to fly souvenirs, coins or things of a similar nature; it will be used for research and the development and manufacture of new products. All kinds of requests have been received. Some people wanted to fly cereal so that they could sell the cereal on the ground as "space" cereal. Not too long ago, a proposal was submitted from a funeral parlor which wanted to spread ashes in space. The request has been turned down. Initially, the Space Shuttle will be used as a research vehicle and to move space application research toward commercial application. But some of the more unusual proposals present some difficult questions. There was a request, for example, from a sculptor who wanted to send up a small package filled with water, which would then be injected into space to form an ice sculpture. In another instance, a musician wanted to go up and play in space to see if it can be done and what effect it might have. There are news broadcasters who want to be the first news broadcasters from space. There are movie producers who want to produce the first movie in space. The most difficult question is who and what should be selected. Some of these requests for the use of the Space Shuttle cannot be rejected lightly.

There has also been a great deal of discussion and research on building assemblies, space stations and space platforms in space. There has been a great deal of discussion

and research about constructing solar power stations in space, converting solar energy into electricity, transmitting it to Earth by microwave or laser and reconverting it on Earth into electricity.

NASA Authority

Insofar as the law is concerned, the most important task with respect to the Shuttle was to determine if there was anything that the Shuttle would do that would be in violation of existing domestic legislation or international treaties and if new domestic legislation was needed to carry out the Shuttle program. More specifically, it had to be determined what authority NASA had under the current Space Act[1] and other applicable statutes and what limitations, if any, might exist that would impact on Space Shuttle operations.

While NASA is a Government agency with a very broad and flexible statute that served us well for over 20 years, we are nonetheless a Government agency whose authority is both granted and limited by statute. NASA cannot, under existing law, run its operation like a commercial operation. NASA is not authorized by statute and recent court decisions to make profit on the services it sells; to set up the normal reserves or revolving funds that industry normally uses to take care of contingencies, risks and the like; or to issue bonds or stock to finance its activities.

For these reasons, NASA assumes no hard contractual obligations for the success or failure of a launch. If something goes wrong with the Shuttle and it has to be brought back or if something went wrong with a payload as a result of something that NASA had done, NASA has no contractual liability for loss or damage to the payload, for loss of revenue, or other consequential damages. There is no guarantee that everything that NASA does will be free of negligence. NASA will use its best efforts, but it does not take on a common carrier type of obligation.

Space Allocation

As to allocation of space on the Shuttle, NASA follows the principle of first-come-first-served. There would be a problem, if one were to auction off space on the Shuttle or allocate it by raising prices. NASA's launch policy, which has been in effect for some time, is based on the principle of nondiscrimination. NASA will not charge anyone more than it charges a domestic commercial concern. This is in line with the provision of the 1967 Outer Space Treaty[2] which states that access to space should be on a nondiscriminatory basis, and if one started auctioning off access to space, the rich nations could buy it and others could not. NASA did not want to decide priorities because if it decided that one commercial customer had priority over another commercial user, it would really have made an economic decision affecting the two entities. Thus, NASA decided that it would follow the first-come-first served principle and not intrude itself into these types of economic decisions. In this connection, the

[1]National Aeronautics and Space Act of 1958, As Amended, and Related Legislation, Pub. L. 85-568, 85th Cong. H.R. 12575 July 29, 1958, 72 Stat. 426, 42 U.S.C. § 2451.

[2][1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967).

suggestion was made to users that if they wished and if they could avoid antitrust implications, they should consider establishing a user's organization that could make decisions on launch priority.

Who Should Operate the Shuttle?

Last year, the National Academy of Engineering undertook a study to determine what sort of operation the Space Shuttle should be; whether it should be retained in government, in NASA or some other government agency—not necessarily NASA; whether it should be a public corporation following the COMSAT model; or whether it should be totally private enterprise, a stock company that would in effect operate the Space Transportation System for all users—government, foreign or private commercial entities. The Academy has not precluded any of these options. It has suggested that at least for now, the Shuttle ought to be a NASA operation.

Insofar as any restriction on a private company buying a Shuttle is concerned, the Outer Space Treaty of 1967[3] stipulates that all national activities shall be under the authorization and continuing supervision of the government. Thus, there must be a government authorization to launch the Shuttle because of the requirement in the Outer Space Treaty. In addition, there is the problem of financing. The Orbiter, once it goes into production, would cost about $100,000,000. In theory, however, there is nothing to prevent a private company from buying a Shuttle except the aforementioned considerations and the need to use existing NASA and Air Force facilities. I doubt these facilities would be turned over for the exclusive use of a private company without Congressional authorization.

Risk, Liability and Insurance

Turning to problems associated with risk, there is commercial insurance available to cover practically every risk in NASA's launch services contract. There is also a provision in the launch services agreement to the effect that the Government agrees not to hold any user liable for damage to the Orbiter and, in return, the user agrees not to hold any other user or the government liable for damage to his particular payload. Thus there are cross waivers in effect, covenants not to sue, as well as an indemnity if a user fails to flow down this provision to any party who could bring a separate action, including an action based on the right of subrogation. This has been done because of a concern about the capacity in the insurance market to insure such a risk. There was also a concern about the mixture of payloads and the differing value of payloads. For instance, a ten thousand dollar payload may explode, causing damage to a $300,000,000 payload. A university, research institute or small business ordinarily would not carry the kind of product or third-party liability insurance that would be needed and, therefore, could not afford to take the risk of flying on the Shuttle.

The past practice has been that commercial users have gone into the market and insured their satellite. They have insured their spacecraft against damage from any cause, based on its stated value (the cost of the spacecraft and associated launch). When COMSAT took out its first policy, the premium was about 14 percent; more recently the

---

[3] *Id.*

premium has come down significantly to about 7 percent. When COMSAT attained its first policy, it had what is called a two-failure deductible; so before you could receive payment on the policy, you had to lose two satellites out of four. This went down to a one-failure deductible policy and then to no deductible at all.

In addition to covering damage to property, the market has been writing loss of revenue insurance, so-called satellite life insurance; so if the satellite stops operating before its anticipated five-year life and loses revenue, you get loss of revenue coverage. Mostly, Lloyds has been involved, but the market has become pretty much an international market with Lloyds participating. When the European Space Agency lost one of its satellites, insurance paid $34,000,000. A more recent loss of an RCA satellite, insured for $70,000,000 with Lloyds, has also been paid.

As to third-party liability insurance, there is a general government policy against the government's buying insurance covering third-party liability risk. Under Section 308a, a recent amendment to the NASA Act[4], NASA has the authority (1) to indemnify (which will take care of the "little" user) and (2) to buy insurance to the extent it is available and needed to protect itself and anybody else flying on the Shuttle. NASA is in the process of issuing regulations implementing this new legislation. In the meantime, NASA requires in its launch services contract with a commercial user or foreign government, that the user provide up to $500,000,000 worth of third-party insurance coverage, or such other amount that may be available, with the U.S. Government being a named insured as well as the particular user. NASA has recently participated in the negotiation of a policy in which the insured is Satellite Business Systems—a partnership of COMSAT, IBM and Aetna Insurance, the United States Government and those prime and subcontractors which the U.S. Government would be required to reimburse for third-party liability losses. There are many provisions in the policy which go beyond the normal commercial aspects of insurance. There is even a provision that any defense based on the sovereign immunity of the United States Government will not be asserted without the express consent of the Government. Thus, the Government could, in a particular situation, waive its sovereign immunity defense under the Federal Tort Claims Act[5]. It also provides for the payment of claims presented under the 1972 Liability Convention[6] and for the U.S. Government to negotiate and control, though in consultation with the underwriters, the settlement of claims.

Insurance availability, third-party liability, property damage or what is called first-party liability and loss of revenue coverage are very important aspects of the ever-expanding commercialization of space. If commercialization is to come about through investment by private enterprise, users will want to spread the risk of loss or liability for damage.

---

[4]42 U.S.C. § 2458 (b).

[5]28 U.S.C. § 1346.

[6]Convention of International Liability for Damage Caused by Space Objects, March 29, 1972, [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (effective Oct. 9, 1973).

New Business Opportunities and Joint Endeavors

Until very recently, all of our commercial users of launch services were the satellite communications companies. However, NASA is committed to involve other sectors of U.S. industry in space-based business opportunities as well.

If researchers in virtually every discipline of science and technology knew that within a hundred miles of their laboratories there existed a facility where they could achieve a near-perfect vacuum, zero gravity and an unlimited vantage point from which to view the Earth, we are convinced that they would jump at the chance to use the laboratory in a host of unique experiments—pharmaceuticals, materials processing, electronics, solid state physics, crystal growth, communications, lasers, biology, geological and geophysical exploration and so on. During the 1980's such a laboratory will be available to U.S. industry, only it will not be a hundred miles down the road; it will be a hundred miles overhead, in Earth orbit. That laboratory, of course, is the Space Shuttle and the European-developed Spacelab.

On June 25, 1979, NASA issued Guidelines Regarding Early Usage of Space for Industrial Purposes. Supplementing those broad guidelines, on August 14, 1979, NASA issued Specific Guidelines Regarding Joint Endeavors with U.S. Domestic Concerns in Materials Processing in Space[7]. And on January 25, 1980, NASA entered into the first such joint endeavor with the McDonnell Douglas Astronautics Company. That endeavor, which involves substantial investment by McDonnell Douglas and its associates, will utilize the unique capabilities of the Space Shuttle to develop and demonstrate the technology of continuous-flow electrophoresis under low gravity conditions. NASA will provide free flight time on the Space Shuttle in return for McDonnell Douglas and its associates agreeing at their expense to conduct a three-phase sequential program involving (1) feasibility studies, planning and ground research and development; (2) flight experimentation and technology development; and (3) applications demonstrations in space. In return for McDonnell Douglas promising to make the results of the endeavor available to the United States public on reasonable terms and conditions, NASA agrees to refrain from entering into a similar joint endeavor or international cooperative agreement directly related to the development of commercial devices and processes which would compete with those expected to result from the McDonnell Douglas effort. NASA is not precluded from selling flight time on the Shuttle to any other organization wanting to conduct the same or similar experiments, but it is precluded from providing free flight time for such experiments. Significantly, NASA will not acquire rights in inventions which may be made by McDonnell Douglas or its associates in the course of the joint endeavor, unless McDonnell Douglas fails to exploit the inventions or terminates the agreement or the NASA Administrator determines that an emergency situation exists. At least two additional proposals for joint endeavors with NASA are now being actively considered.

Conclusion

Insurance to reduce the risk, incentives that NASA is willing to make available to U.S. industry, and the demonstrated willingness of industry and the financial

[7] 44 Fed. Reg. 47, 650 (1979).

community to invest their funds in space ventures make it clear to me that the new Shuttle capabilities will exponentially increase commercial activities in space during the decade of the eighties. With the year 2000 only 20 years away, the next century may very well see large numbers of people working and living in space, producing new and improved products for those who have to remain here on Earth.